MA 324 | END-SEMESTER EXAMINATION
STATISTICAL INFERENCE AND MULTIVARIATE ANALYSIS | 14:00–17:00 IST
IIT GUWAHATI | MAY 8, 2026

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	1	9	15

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$\begin{aligned} Y_1 &= \theta_1 + \theta_2 + \epsilon_1, \\ Y_2 &= \theta_1 - 2\theta_2 + \epsilon_2, \\ Y_3 &= 2\theta_1 - \theta_2 + \epsilon_3, \end{aligned}$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2.

$y_i = x_{i1}\theta_1 + x_{i2}\theta_2 + \epsilon_i$

we want to minimise $\sum_i (y_i - x_{i1}\theta_1 - x_{i2}\theta_2)^2$

diff wrt $\theta_1 \Rightarrow \sum_i 2(y_i - x_{i1}\theta_1 - x_{i2}\theta_2)(-x_{i1}) = 0$

$= \sum_i (y_i - x_{i1}\theta_1 - x_{i2}\theta_2) x_{i1} = 0$ ——— ①

diff wrt $\theta_2 \Rightarrow \sum_i (y_i - x_{i1}\theta_1 - x_{i2}\theta_2)(x_{i2}) = 0$ ——— ②

① $\Rightarrow \sum (x_{i1}y_i - x_{i1}^2\theta_1 - x_{i2}x_{i1}\theta_2) = 0$

② $\Rightarrow \sum (x_{i2}y_i - x_{i2}^2\theta_2 - x_{i2}x_{i1}\theta_1) = 0$

we want to minimise $(y_1 - \theta_1 - \theta_2)^2 + (y_2 - \theta_1 + 2\theta_2)^2 + (y_3 - 2\theta_1 + \theta_2)^2$ wrt θ_1, θ_2

diff wrt $\theta_1 \Rightarrow 2(y_1 - \theta_1 - \theta_2)(-1) + 2(y_2 - \theta_1 + 2\theta_2)(-1) + (y_3 - 2\theta_1 + \theta_2)(2)(-2) = 0$

$y_1 - \theta_1 - \theta_2 + y_2 - \theta_1 + 2\theta_2 + 2y_3 - 4\theta_1 + \theta_2 \times 2 = 0$

$\theta_1(-1 - 1 - 4) + \theta_2(-1 + 2 + 2) + (y_1 + y_2 + 2y_3) = 0$

$-6\theta_1 + 3\theta_2 + (y_1 + y_2 + 2y_3) = 0$ ——— ①

Diff wrt $\theta_2 \Rightarrow 2(y_1 - \theta_1 - \theta_2)(-1) + 2(y_2 - \theta_1 + 2\theta_2)(2) + 2(y_3 - 2\theta_1 + \theta_2) = 0$

$-y_1 + \theta_1 + \theta_2 + 2y_2 - 2\theta_1 + 4\theta_2 + y_3 - 2\theta_1 + \theta_2 = 0$

$\theta_1(+1 - 2 - 2) + \theta_2(+1 + 4 + 1) + (-y_1 + 2y_2 + y_3) = 0$

$-3\theta_1 + 6\theta_2 + (-y_1 + 2y_2 + y_3) = 0$ ——— ②

upon solving ① and ② for θ_1 and θ_2 we get the LSE for θ_1 and θ_2

$$-6\theta_1 + 3\theta_2 + (y_1 + y_2 + 2y_3) = 0 \quad \text{—— ①}$$

$$-3\theta_1 + 6\theta_2 + (-y_1 + 2y_2 + y_3) = 0 \quad \text{—— ②}$$

$$-6\theta_1 + 12\theta_2 + (-2y_1 + 4y_2 + 2y_3) = 0 \quad \text{—— ③} \quad (② \times 2)$$

③ - ① $\Rightarrow$ $9\theta_2 + (-3y_1 + 3y_2) = 0$

$$\theta_2 = \frac{3y_1 - 3y_2}{9} = \frac{y_1 - y_2}{3} \Rightarrow \boxed{\hat{\theta}_2 = \frac{y_1 - y_2}{3}}$$

$$-12\theta_1 + 6\theta_2 + 2y_1 + 2y_2 + 4y_3 = 0 \quad \text{—— ④} \quad (1 \times ②)$$

② + -④

$$9\theta_1 - 3y_1 - 3y_3 = 0$$

$$\boxed{\hat{\theta}_1 = \frac{y_1 + y_3}{3}}$$

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\boldsymbol{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
   (a) (3 points) Compute the correlation matrix $\boldsymbol{\rho}$ of $\boldsymbol{X}$.
   (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

a) let U be uniformly distributed on (0,1) then $E(U) = \mu$

$var(u) = \sigma^2$

$cov(X_1, X_2) = cov(U_1, U_2) = 0$ ($\because$ they are independent)

$cov(X_1, X_3) = cov(U_1, U_1+U_2) = cov(U_1, U_1) + cov(U_1, U_2) = var(U_1) = \sigma^2$

$cov(X_2, X_3) = cov(U_2, U_1+U_2) = cov(U_1, U_2) + cov(U_2, U_2) = Var(U_2) = \sigma^2$

variance covariance matrix $=$

$$\begin{array}{c|ccc} & X_1 & X_2 & X_3 \\ \hline X_1 & \sigma^2 & 0 & \sigma^2 \\ X_2 & 0 & \sigma^2 & \sigma^2 \\ X_3 & \sigma^2 & \sigma^2 & \sigma^2 \end{array}$$

correlation btw (X_1, X_2) $= \dfrac{cov(X_1, X_2)}{\sqrt{var(X_1)}\sqrt{var X_2}}$

$\rho(X_1, X_2) = 0/\sigma \times \sigma = 0$ $\qquad \rho(X_2, X_3) = \sigma^2/\sigma \times \sigma = 1$

$\rho(X_1, X_3) = \sigma^2/\sigma \times \sigma = 1$

correlation matrix ρ of X $=$

$$\begin{array}{c|ccc} & X_1 & X_2 & X_3 \\ \hline X_1 & 1 & 0 & 1 \\ X_2 & 0 & 1 & 1 \\ X_3 & 1 & 1 & 1 \end{array}$$

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b) to find principal components we need eigen vectors and eigen values of ρ

$$|\rho - I\lambda| = 0 \Rightarrow \det\begin{bmatrix} 1-\lambda & 0 & 1 \\ 0 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{bmatrix} = 0$$

$$(1-\lambda)\left((1-\lambda)^2 - 1\right) + 1\left(-(1-\lambda)\right) = 0 \Rightarrow (1-\lambda)^3 - (1-\lambda) - (1-\lambda) = 0$$

$$(1-\lambda)^3 - 2(1-\lambda) = 0$$

$$(1-\lambda)\left((1-\lambda)^2 - 2\right) = 0$$

$$(1-\lambda)(1+\lambda^2-2\lambda-2) = 0 \Rightarrow (1-\lambda)(\lambda^2-2\lambda-2) = 0$$

① $\lambda = 1$ ② $(1-\lambda)^2 - 2 = 0$

$$\Rightarrow (1-\lambda)^2 = 2$$

$$= (1-\lambda) = \pm\sqrt{2}$$

a) $\lambda = 1-\sqrt{2}$ b) $\lambda = 1+\sqrt{2}$

Eigen ~~value~~ vector for ① i.e $\lambda = 1$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{12} \\ e_{13} \end{bmatrix} = 0 \Rightarrow e_{13} = 0 \Rightarrow e_{11} + e_{12} = 0$$

$$\Rightarrow \cancel{\begin{bmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}} \quad \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix} = e_1$$

Eigen vector for ② a) i.e $\lambda = 1-\sqrt{2}$

$$\begin{bmatrix} \sqrt{2} & 0 & 1 \\ 0 & \sqrt{2} & 1 \\ 1 & 1 & \sqrt{2} \end{bmatrix} \begin{bmatrix} e_{21} \\ e_{22} \\ e_{23} \end{bmatrix} = 0 \Rightarrow \left.\begin{array}{l} \sqrt{2}e_{21} + e_{23} = 0 \\ \sqrt{2}e_{22} + e_{23} = 0 \\ e_{21} + e_{21} + \sqrt{2}e_{23} = 0 \end{array}\right\} \Rightarrow \begin{array}{l} e_{21} = e_{22} \\ e_{23} = -\sqrt{2}e_{21} \end{array}$$

$$\Rightarrow \begin{bmatrix} 1/2 \\ 1/2 \\ -\frac{\sqrt{2}}{2} \end{bmatrix} = e_2$$

Eigen vector ② (b) i.e $\lambda = 1+\sqrt{2}$

$$\begin{bmatrix} -\sqrt{2} & 0 & 1 \\ 0 & -\sqrt{2} & 1 \\ 1 & 1 & -\sqrt{2} \end{bmatrix} \begin{bmatrix} e_{31} \\ e_{32} \\ e_{33} \end{bmatrix} \Rightarrow \left.\begin{array}{l} -\sqrt{2}e_{31} + e_{33} = 0 \\ -\sqrt{2}e_{32} + e_{33} = 0 \\ e_{31} + e_{32} + -\sqrt{2}e_{33} = 0 \end{array}\right\} \begin{array}{l} e_{31} = e_{32} \\ e_{33} = \sqrt{2}e_{32} \end{array}$$

$$\Rightarrow \cancel{\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}} \quad \begin{bmatrix} 1/2 \\ 1/2 \\ \sqrt{2}/2 \end{bmatrix} = e_3$$

$$Z = \begin{bmatrix} \frac{X_1-\mu_1}{\sigma_1} \\ \frac{X_2-\mu_2}{\sigma_2} \\ \frac{X_3-\mu_3}{\sigma_3} \end{bmatrix} = \begin{bmatrix} \frac{X_1-\mu}{\sigma^2} \\ (X_2-\mu)/\sigma^2 \\ \frac{X_3-2\mu}{\sigma^2} \end{bmatrix}$$

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continued in the end!!

3. Let the matrix of distance be

$$\begin{array}{c|c} & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{array}$$

(a) (6 points) Cluster the four items using each of the following procedures:

i. Complete linkage hierarchical procedure. -

ii. Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

i) compleute $d_{uv} \Rightarrow$ max distance ii) avegare $\Rightarrow$ we take avg of all the dist

(i) ① Based on above distance matrix 1,2 have min dist hence we combine

$$\begin{array}{c|c} & \begin{array}{ccc} (1,2) & 3 & 4 \end{array} \\ \begin{array}{c} (1,2) \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & & \\ 11 & 0 & \\ 5 & 3.5 & 0 \end{bmatrix} \end{array}$$

$(4,(1,2)) = \max((4,1),(4,2))$

$= \max(5,3) = 5$

$(3,(1,2)) = \max((3,1),(3,2))$

$= \max(11,2) = 11$

② we notice that (3,4) have min dist, hence we combine (3,4)

$$\begin{array}{c|c} & \begin{array}{cc} (1,2) & (3,4) \end{array} \\ \begin{array}{c} (1,2) \\ (3,4) \end{array} & \begin{bmatrix} 0 & \\ 11 & 0 \end{bmatrix} \end{array}$$

$((1,2),(3,4)) = \max\left(((1,2),3),((1,2),4)\right)$

$= \max(11,5) = 11$

③ Now olny 2 clusters remain, hence we combine them.

(1,2,3,4)

(ii) ① Based one distance matrix given in question 1,2 have min dist hence we combine (1,2)

$$\begin{array}{c|c} & \begin{array}{ccc} (1,2) & 3 & 4 \end{array} \\ \begin{array}{c} (1,2) \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & & \\ 6.5 & 0 & \\ 4 & 3.5 & 0 \end{bmatrix} \end{array}$$

$(4,(1,2)) = \dfrac{(4,1)+(4,2)}{2} = \dfrac{5+3}{2} = \dfrac{8}{2} = 4$

$(3,(1,2)) = \dfrac{(3,1),(3,2)}{2} = \dfrac{11+2}{2} = \dfrac{13}{2} = 6.5$

(2) Based on the new dist matrix (3, 4) are the closest hence we combine them

$$\begin{array}{c|cc} & (1,2) & (3,4) \\ \hline (1,2) & 0 & \\ (3,4) & 5.25 & 0 \end{array}$$

$$((1,2),(3,4)) = \frac{(1,3)+(1,4)+(2,3)+(2,4)}{4}$$

$$= \frac{11+5+2+3}{4} = \frac{21}{4} = 5.25$$

(3) Now only 2 clusters remain hence we combine them, to get (1,2,3,4)

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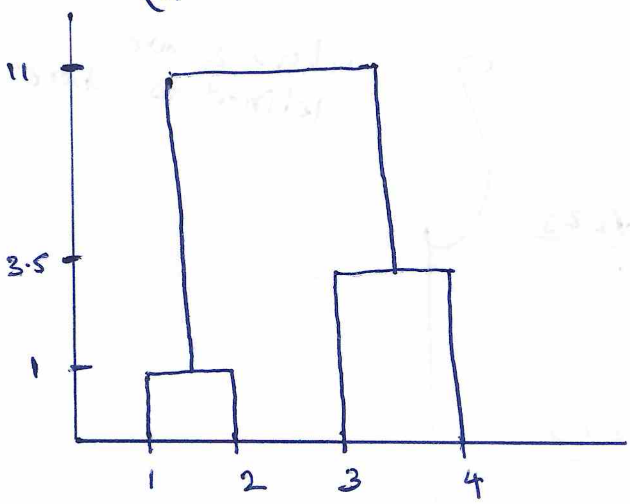
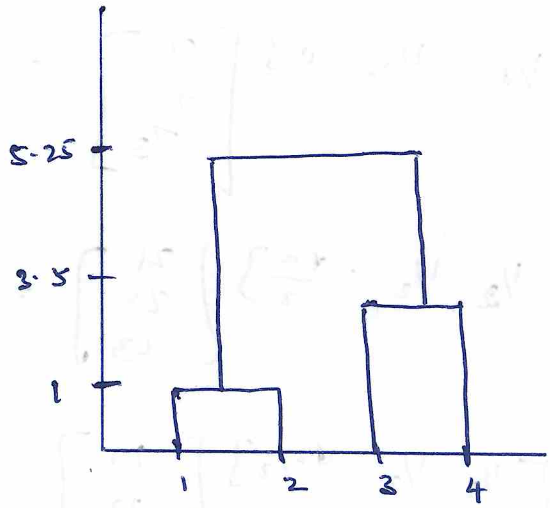
b) DENDRO DIAGRAM

(i)



ii)



The clusters that are formed from both the methods are same and the level (ie distance) at which we link them is also same for the 1st clusters formed i-e (1,2) and (3,4) the mainly differ dete the level at which we form the 3rd cluster. 1st method at level 11 2nd method at level 5.25.

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ADDITIONAL PAGES FOR ANSWERS

2) b)

1) $\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{x_1}{\sqrt{2}} - \frac{x_2}{\sqrt{2}}$

② $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{x_1}{2} + \frac{x_2}{2} - \frac{\sqrt{2}x_3}{2}$

③ $\begin{bmatrix} 1/2 & 1/2 & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{x_1}{2} + \frac{x_2}{2} + \frac{\sqrt{2}x_3}{2}$

① $\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \frac{z_1}{\sqrt{2}} - \frac{z_2}{\sqrt{2}}$

$\begin{bmatrix} 1/2 & 1/2 & -\frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \frac{z_1 + z_2 - \sqrt{2}z_3}{2}$

$\begin{bmatrix} 1/2 & 1/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \frac{z_1 + z_2 + \sqrt{2}z_3}{2}$

where z are defined as above

$\boxed{\text{variance of 1st principal componed max eigen value} \Rightarrow 1+\sqrt{2}}$

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